

1. Executive Summary

		Traffic Summary		Count	
		Total Flows Observed		131211	
Total Threats Mitigated		18		5200829172	
High Priority		9		5189881328	
Low Priority		9		95903	
Manual Releases		0			
Manual Blocks		0			
ML Metrics		Value		Attack Vector	
False Positives		258		ICMP Flood	
FP Rate		0.23%		SYN Flood	
IF Precision		99.38%		UDP Flood	
IF Recall		96.93%		Uncertain	
IF F1-Score		98.14%			
IF Accuracy		98.99%			
				Action	
				Quarantined	
				Time Ban	
				Blackhole	
				Released	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.38%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	96.93%	Share of actual attacks successfully flagged
F1-Score	98.14%	Balanced measure combining Precision and Recall
Accuracy	98.99%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.23%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	3.07%	Actual attacks that went undetected
True Positive Rate (TPR)	96.93%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.77%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 34917	FN 1107
Actual: Normal	FP 219	TN 94796

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	34917	0
Act: UDP	0	0	0

2c. Response Latency

Metric	Value	Description
Detection Time	287.6 ms	Average time taken to detect an attack
Mitigation Response Time	5.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.00%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.50%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	3.87%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.23	3	3	2026-09-15 12:25:14	2026-09-15 12:26:36
10.0.0.24	2	2	2026-09-15 12:25:14	2026-09-15 12:26:24
10.0.0.25	2	2	2026-09-15 12:25:17	2026-09-15 12:26:28
10.0.0.1	4	1	2026-09-15 12:25:52	2026-09-15 12:26:56

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 11:41:23	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 11:41:24	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 11:41:26	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 11:41:33	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-15 11:41:33	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-15 11:41:34	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-15 11:41:34	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-15 11:41:36	10.0.0.24	DDoS	ICMP Flood	91.3%	Critical	Time Ban (1m)
2026-09-15 11:41:36	10.0.0.24	DDoS	ICMP Flood	91.3%	Critical	Time Ban
2026-09-15 11:42:33	10.0.0.23	DDoS	ICMP Flood	92.1%	High	Time Ban (2m)
2026-09-15 11:42:34	10.0.0.25	DDoS	ICMP Flood	94.8%	High	Time Ban (2m)
2026-09-15 11:42:36	10.0.0.24	DDoS	ICMP Flood	92.1%	High	Time Ban (2m)
2026-09-15 11:48:17	10.0.0.7	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-15 11:48:29	10.0.0.24	DDoS	ICMP Flood	92.1%	High	Released
2026-09-15 11:58:20	10.0.0.4	DDoS	Uncertain	58.6%	Low	Released
2026-09-15 12:03:49	10.0.0.23	DDoS	ICMP Flood	92.7%	High	Released
2026-09-15 12:10:19	10.0.0.25	DDoS	ICMP Flood	94.6%	High	Released
2026-09-15 12:26:02	10.0.0.1	DDoS	Uncertain	59.8%	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.1	Uncertain	0.6097	59.8%	Low	Phase 1	1	11s	2026-09-15 12:26:56	Attack Stopped
10.0.0.1	Uncertain	0.6158	59.8%	Low	Phase 1	1	11s	2026-09-15 12:26:45	Attack Stopped
10.0.0.23	ICMP Flood	0.6356	94.1%	Critical	Phase 2	3	11s	2026-09-15 12:26:36	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.6134	59.8%	Low	Phase 1	1	10s	2026-09-15 12:26:34	Attack Stopped
10.0.0.25	ICMP Flood	0.6416	92.1%	Critical	Phase 2	2	1m 11s	2026-09-15 12:26:28	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6336	94.1%	High	Phase 2	1	1m 11s	2026-09-15 12:26:25	Ban Expired
10.0.0.24	ICMP Flood	0.6419	92.1%	High	Phase 2	2	1m 10s	2026-09-15 12:26:24	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.6165	59.8%	Low	Phase 1	1	10s	2026-09-15 12:26:02	Attack Stopped
10.0.0.25	ICMP Flood	0.6444	91.2%	High	Phase 2	1	11s	2026-09-15 12:25:28	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.23	ICMP Flood	0.6454	91.1%	High	Phase 2	1	11s	2026-09-15 12:25:25	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6455	91.0%	High	Phase 2	1	10s	2026-09-15 12:25:24	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE